

Assignment 2: Fine-Tuning a Language Model for an NLP Problem

Overview

In this assignment, you will individually design and implement an NLP system to address a classification task of your choice. You will fine-tune **either an LSTM-based model or a BERT-based model** on your chosen dataset, then evaluate and discuss the results.

Scope & Objectives

1. Choice of Problem Statement

- You are free to select any classification problem. Possible examples include:
 - **News classification for credit sentiment** (e.g., positive or negative credit signals).
 - **News classification for liquidity direction** (e.g., is liquidity increasing or decreasing?).
 - **News sentiment classification for the forex market** (e.g., determining sentiment polarity in relation to currency movements).
 - **Classifying tweets on crypto coins** (e.g., bullish vs. bearish sentiment).
- The chosen problem must be well-defined, with a clear classification objective (binary or multi-class).

2. Data Sourcing & Preparation

- **Data Sourcing:** You are responsible for finding data relevant to the problem you propose. You may obtain data from:
 - Public datasets (e.g., **Kaggle**, **Hugging Face Datasets**, etc.)
 - Web scraping (ensure you respect data usage policies)
 - Other publicly available sources
- **Data Preparation:** Perform all necessary data cleaning, preprocessing, and labeling. Clearly document any decisions or transformations made.

3. Model Choice & Fine-Tuning

- Choose between **LSTM** and **BERT** (or variants like FinBERT, RoBERTa).
- Provide a brief rationale for your chosen model in the context of your financial classification task.
- Use appropriate **fine-tuning** techniques for your model.

4. Implementation, Evaluation & Discussion

- Provide an **end-to-end workflow** covering:
 - Data preprocessing and feature engineering
 - Model architecture and training/fine-tuning
 - Model evaluation
- Evaluate model performance using relevant metrics (e.g., accuracy, precision, recall, F1-score, or AUC).
- Summarize and discuss:
 - Data challenges and preprocessing steps
 - Model hyperparameters and architecture choices
 - Obstacles encountered and solutions employed
 - Potential improvements or future directions

Deliverables

1. Final Submission Zip File

- Your final submission should be **one zip file**. The zip file shall include a **Jupyter Notebook (.ipynb)** and **the exported pdf file** containing:
 - **Code**: End-to-end implementation (data loading, preprocessing, model fine-tuning, and evaluation).
 - **Explanations/Comments**: Clear documentation for each step, describing your reasoning and methodology.
 - **Model Evaluation**: Include performance metrics and any relevant plots or tables for clarity.
 - **Results Discussion**: Discuss key findings, analyze the performance, and suggest possible improvements.

Assessment Criteria

1. Novelty of Financial Problem Statement

- Creativity and specificity of the selected classification task.
- Relevance to financial applications and justification.

2. Effort in Data Preparation

- Thoroughness in sourcing and cleaning the data.
- Proper handling of data preprocessing steps.

3. Code Quality & Clarity

- Readability, logical structure, and maintainability of the code.
- Inclusion of sufficient comments and explanations.

4. End-to-End Implementation

- Completeness of the workflow: data processing, model design, training, and evaluation.
- Reproducibility: Notebook should run seamlessly given access to the required data.

5. Discussion & Critical Analysis

- Evaluation and interpretation of results.
- Reflection on strengths, weaknesses, and potential future work.

Important Notes

- **Individual Work:** This is not a group assignment; all work should reflect your own efforts.
- **Plagiarism Policy:** You must adhere to academic integrity standards. Any form of plagiarism will result in severe penalties.
- **Deadline:** You need to submit your work by **April 6, 23:59**. Late submissions will incur penalties.